ETHNOHACK 2026 · TRACK 2 · NATURAL PRODUCTS FOR DRUG DISCOVERY

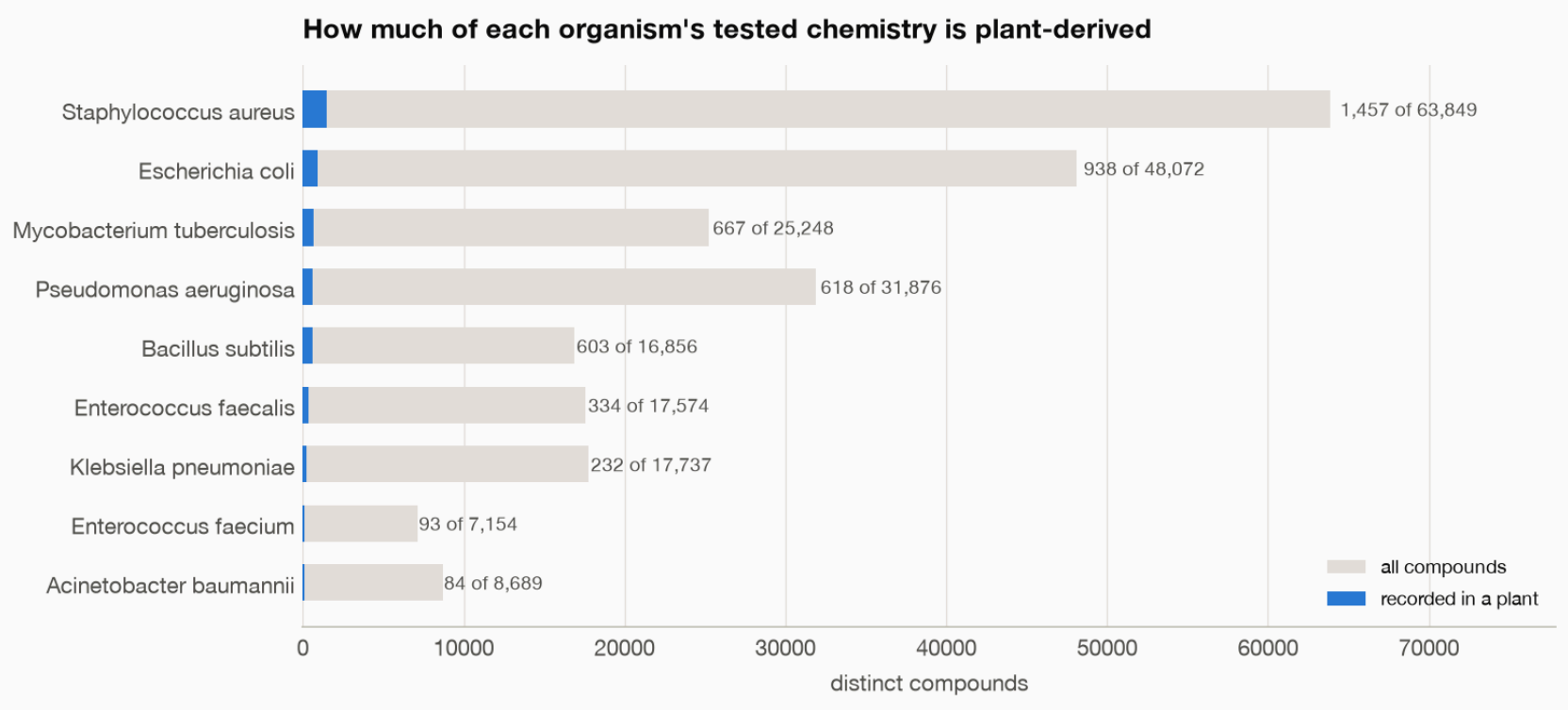
Among hundreds of thousands of plant compounds, which deserve a closer look?

We took the track question literally: three bacterial pathogens, one model each, a ranked shortlist of plant-derived compounds for the bench — and a measured account of exactly where that shortlist stops being evidence.

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Everything shown is a model prediction over published laboratory measurements. Nothing here is a clinical claim, a statement of efficacy, or medical advice.

Medicine has measured a great deal of chemistry against these organisms. Almost none of it came from a plant.



ChEMBL MIC pull, before curation. After curation, *S. aureus* has 1,424 plant-derived compounds out of 58,191 — 2.4%. Meanwhile 97,709 plant structures with a known source organism carry no measurement against any of these three at all.

You cannot look up an experiment nobody has run. That gap is the entire reason to predict rather than retrieve.

Three pathogens, three different cell envelopes, one model each.

We planned per-protein models. Then we measured the download, and changed the project to fit the data rather than the other way round.

99.6%

of MIC rows for these organisms are measured against the **whole organism**, not a single protein

35

single-protein targets carry any MIC at all, median 8 compounds — too few to train on

97,709

unmeasured plant structures screened, against 58,191 / 43,099 / 22,907 labelled compounds

THE THREE ENVELOPES

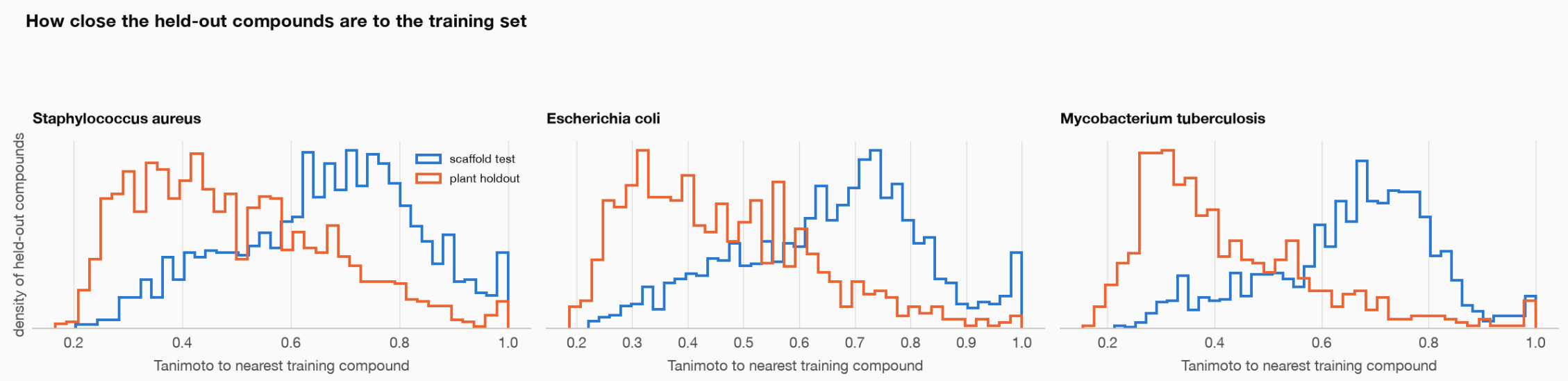
- *Staphylococcus aureus* — Gram-positive
- *Escherichia coli* — Gram-negative
- *Mycobacterium tuberculosis* — acid-fast, the hardest wall and the least data

WHAT THE MODEL THEREFORE CLAIMS

Whether a compound stops a given bacterium growing. It says nothing about mechanism, and we do not let it pretend otherwise. The protein layer annotates a shortlist; it never ranks one.

Holding out chemical scaffolds is not enough. We hold out every plant-derived compound.

We cluster first and split the clusters, so no compound and no chemical scaffold appears on both sides of a split. The model cannot pass by having nearly seen the answer.



Removing shared ring skeletons removes none of the resemblance that matters. The median held-out **scaffold-test** compound still has a close neighbour in training (*S. aureus*: Tanimoto `0.680`); the median **plant** compound sits far further out (`0.461`). Only `3.4%` of plant compounds have a near-twin in training, against `13.5%` of scaffold-test compounds.

That distance is the whole problem — and it is why a nearest-training Tanimoto is printed next to every single score we ship.

AUROC falls on the plant holdout for every pathogen. That is the number we rank on.

PATHOGEN	VALIDATION	SCAFFOLD TEST	PLANT HOLDOUT	95% CI	PLANT BASE RATE	P@10%	ENRICHMENT, TOP 5%
Staphylococcus aureus	0.880	0.863	0.719	0.689–0.749	0.253	0.570	2.45x
Escherichia coli	0.882	0.903	0.682	0.609–0.750	0.090	0.299	5.03x
Mycobacterium tuberculosis	0.846	0.815	0.629	0.582–0.677	0.237	0.435	2.29x

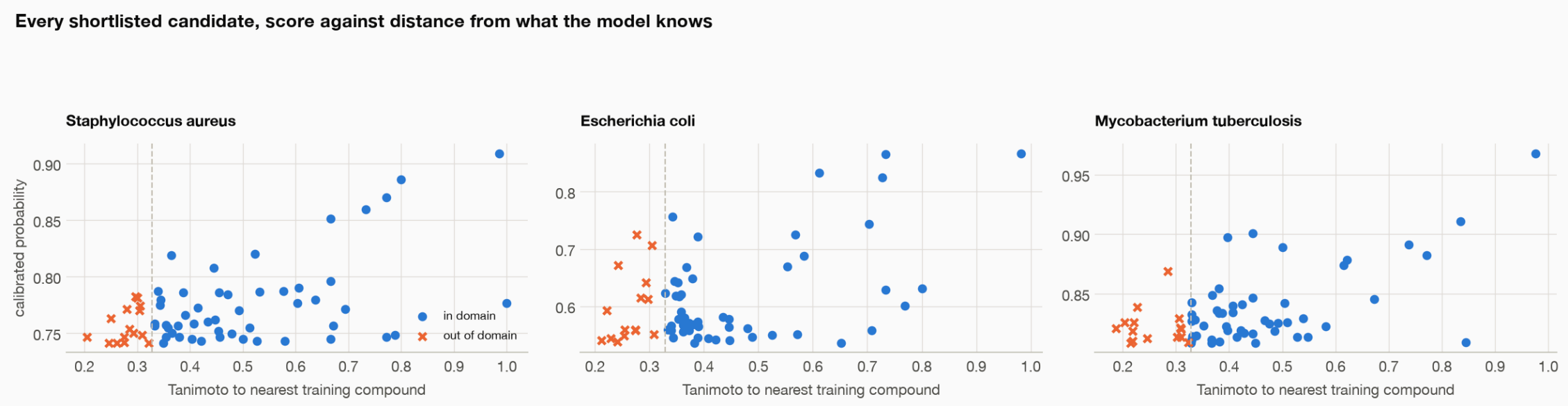
WHAT THIS BUYS A LAB

Precision at the top of the list is what a chemist spends money against. On *S. aureus*, 57% of the top tenth are active against a 25% base rate. On *E. coli*, five times the hit rate of picking at random. The claim is not a cure. It is a shorter queue for the bench.

ONE THING WE WILL NOT SMOOTH OVER

These rungs are **not** a tidy descending staircase. *E. coli* scores `higher` on the scaffold test (0.903) than on validation (0.882). The claim we can defend is narrower and still the one that matters: **the plant holdout is consistently the hardest of the three, for every pathogen** — and it is the only one that measures what a plant screen actually asks.

Every score travels with its distance from anything the model has seen.



The dashed line is the applicability domain at Tanimoto `0.328` — not chosen for looks, but the 25th percentile of the plant holdout, the most distant set on which we actually measured the model. Below it, candidates are struck through: we publish them in a separate tier that states in its own words that its scores are not evidence of activity. We do not rank them.

Per pathogen: 50 in-domain rows and 15 exploratory, each with predicted score, nearest-training Tanimoto, source plant and family, and a literature DOI.

The chemistry of each shortlist tracks the cell envelope it was selected against.

The model was given fingerprints and physicochemical descriptors. Nobody told it about cell walls.

30 / 50

M. tuberculosis, acid-fast — alkaloids (17) and terpenoids (13)

21 / 50

E. coli, Gram-negative — peptides, sugars and fatty acids

19 / 50

S. aureus, Gram-positive — shikimates and phenylpropanoids

> We report this as a **hypothesis worth testing, not a finding**. It partly reflects which compounds LOTUS has annotated, and three shortlists of fifty are not enough to establish it. It is on this slide because it is the most interesting thing the screen produced, and because the way to settle it is to go and look.

The thing we set out to show did not replicate. We are reporting it anyway.

The hypothesis: holding out whole plant **families** should be harder than holding out chemical **scaffolds**, because families share biosynthesis. We measured the gap four separate times, in a standalone experiment on 400–823 compounds — smaller than, and separate from, the main pipeline above.

MEASUREMENT	FAMILY – SCAFFOLD	PAIRED SD	FAMILY WORSE IN	VERDICT
S. aureus	−0.073	0.067	8 of 10	direction holds
E. coli	+0.004	0.139	2 of 10	reverses
M. tuberculosis	−0.097	0.158	7 of 10	spread exceeds effect
Earlier exploratory run	−0.071	–	–	not reproduced at scale

The direction is usually negative; the magnitude never settles, and on *E. coli* it reverses outright. Two of the three estimates rest on test sets of roughly a hundred compounds, which is the likely reason they say so little. **So it goes in as an honest negative, not as our contribution** — and the cluster-level scaffold split, which does replicate cleanly on all three organisms, is what carries the evaluation on the previous slides.

The part we would rather a judge heard from us than found alone.

DEFECTS IN THE SHIPPED SHORTLISTS

- The **top-ranked candidate in all three** is a near-copy of a training compound — Tanimoto `0.986` / `0.982` / `0.976`. In two of the three it is the same non-plant polyketide, with no source plant and no DOI.
- One row ships at Tanimoto `1.000`, correctly flagged as a fingerprint twin of a training compound.
- Missing plant provenance on 12 / 8 / 3 of 50 rows. DOI coverage 38 / 42 / 47 of 50.
- Per-compound attributions came out numerically impossible, so we withhold that column rather than show it.

LIMITS OF THE METHOD ITSELF

- Labels carry publication bias: most published MIC values sit at or below the active threshold, because inactive compounds are reported less often.
- No mechanism. A whole-organism readout cannot tell you what a compound binds.
- Model and shortlist stages have **not** been independently reproduced; curation and splits have.
- No experimental validation of any kind. Not one of these compounds has been put on a plate by us.

> A screen that only reports its wins is not a result.

Every number on the site and in this deck is read from a committed result file.

57

ledger metrics re-run end to end from the raw downloads on a second machine — all matched to the last decimal

12

pipeline stages, each writing its config hash and metrics to `results/runs.jsonl`

0

numbers typed by hand into the web app; it reads `results/` at build time and retrains nothing

DATA AND LICENCES

- **ChEMBL** (CC BY-SA) — the MIC activity labels, whole-organism.
- **LOTUS** (CC0) — the plant-to-compound edge, with references.
- **GBIF** — cross-check of which source organisms are plants.
- **NPASS** was evaluated and **dropped**: its MIC table is a copy of ChEMBL, so we cite ChEMBL directly.

WHAT IS NEXT

- The protein layer, from the IC50 and Ki download that already exists.
- **The honest next step is not a fourth pathogen.** It is putting ten of these on a real plate and publishing the hit rate next to the predicted one — including if it is worse.

> The plant records carry traditional knowledge. Anything built on this carries the Nagoya Protocol and Local Contexts with it as a condition, not a footnote. Generative-AI assistance is disclosed in the repository README.